INCH POUND

MIL-C-3098/52E
10 April 1995
SUPERSEDING
MIL-C-3098/52D
4 October 1976

MILITARY SPECIFICATION SHEET

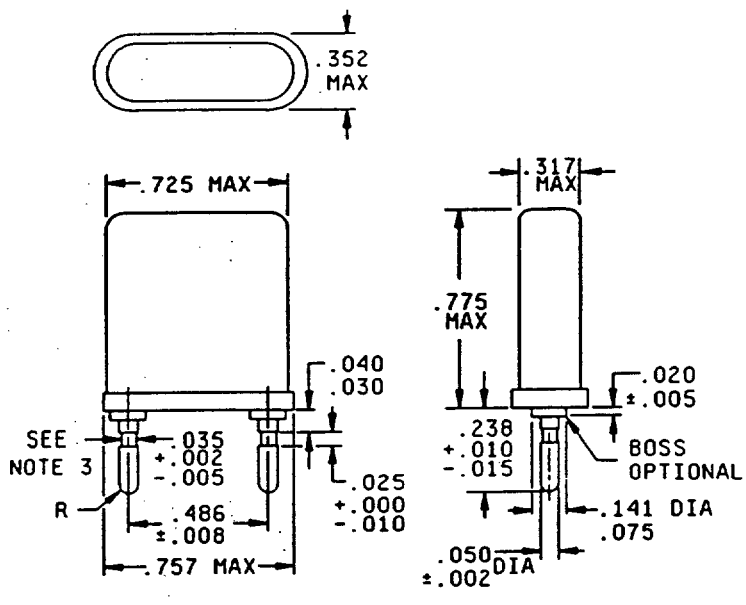
# CRYSTAL UNIT, QUARTZ, CR-75/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

**Pertinent characteristics: 50.0 MHz to 125 MHz; fifth overtone; controlled; series.**

| Inches | mm |
|---|---|
| .002 | 0.05 |
| .005 | 0.13 |
| .008 | 0.20 |
| .010 | 0.25 |
| .015 | 0.38 |
| .020 | 0.51 |
| .025 | 0.64 |
| .030 | 0.76 |
| .035 | 0.89 |
| .040 | 1.02 |
| .050 | 1.27 |
| .075 | 1.91 |
| .141 | 3.58 |
| .238 | 6.05 |
| .317 | 8.05 |
| .352 | 8.94 |
| .486 | 12.34 |
| .725 | 18.42 |
| .757 | 19.23 |
| .775 | 19.69 |




NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR-75/U.


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REQUIREMENTS:

Holder: HC-6/U (MIL-H-10056/2) or HC-36/U (MIL-H-10056/23) or HC-48/U (MIL-H-10056/35).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 50.0 MHz to 125 MHz, inclusive.

Frequency tolerance:

Operating temperature range: ±10 ppm.

Room temperature: ±70 ppm.

Equivalent resistance: ±40 ohms, maximum.

Mode of oscillation: Fifth mechanical overtone.

Reference temperature: 75°C ±1°C.

Temperature ranges:

Operable temperature range: -55°C to +70°C and +80°C to +90°C, inclusive.

Operating temperature range: +70°C to +80°C, inclusive.

Drive level: 1.0 ±0.2 mW.

Capacitance, shunt: 7.0 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: MIL-STD-202, method 201.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

NOTES:
1. The socket designed to mate with crystal holder HC-6/U is designated in MIL-S-12883/04-01 (see MIL-S-12883/4). For Navy use, other sockets listed in MIL-STD-242 may be used.
2. NATO preferred type NXT55.

Revision letters are not used to denote changes due to the extensiveness of the changes.



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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